Paraglider flight logs as hypsometers: daily temperature profiles of an Alpine valley, its diurnal heating, and the nearly level ceiling of thermals

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Abstract

A paraglider flight recorder stores both pressure and GNSS altitude: together they turn routine flight logs into hypsometric measurements of layer-mean virtual temperature. From 86 369 IGC logs of the Valais, Swiss Alps, quality screening retains 13 520 dedicated-instrument flights on 437 days in 2011–2026; 392 days provide at least one profile layer supported by ten flights. At midday, between about 2200 and 3600 m, the profiles differ from the Payerne radiosonde, 100 km away, by a median of $+0.67$ K with a scaled median absolute deviation of 1.47 K (1545 layer comparisons on 370 days). Between morning (07–10 UTC) and afternoon (14–18 UTC), temperatures increase by 3.6 K near 2200 m (95 % interval 3.0–4.1 K) and 1.7 K near 3020 m (1.3–2.2 K). Hourly estimates resolve the daytime increase, and pointwise removal of fixes within 300 m of terrain preserves it. On matched observing days, median changes in the humidity contribution at Payerne are only a few tenths of a kelvin, supporting a predominantly thermal interpretation of the multi-kelvin valley signal. The upper envelope of 53 320 observed climbs follows terrain by factors of only 0.2–0.3 at scales of 2.5–10 km. A hydrostatic thought experiment explains how flight geometry can suppress a local warm-slope signal and motivates a comparison of slope-following and slope-leaving climbs. Routine flight archives thus provide a repeatable, vertically resolved record of daytime valley warming without additional airborne sensors.

Keywords: paraglider flight logs; hypsometer; temperature profiles; thermals; Alpine valley; convective boundary layer; mountain meteorology; crowdsourced observations

1 Introduction

Recreational paragliders sample the convective boundary layer of the Alps for hours on many flying days. Their logs, uploaded to online contests, provide a spatially extensive but selective record: weather, pilot decisions and flight access determine where and when observations exist. A companion paper (Korner, 2026b) read the vertical motion of the air from these logs, after deriving the sink polars of the gliders from the logs themselves. This paper reads their temperature, with a method that needs nothing beyond what the logs contain, tests it against a radiosonde on hundreds of days, and uses it for two questions about the mountain boundary layer on which campaigns and models disagree.

The convective boundary layer over mountains is poorly observed (Rotach and Zardi, 2007; Serafin et al., 2018; Lehner and Rotach, 2018). Routine radiosonde networks provide limited coverage inside mountain valleys, while dedicated campaigns resolve processes over shorter periods. The relationship between local thermal structure and the valley-scale boundary

layer remains an active question. De Wekker and Kossmann (2015), reviewing the concepts, describe a boundary-layer top that follows the terrain in the morning and may become level in the afternoon when the valleys heat fast enough, and note that detailed observations of this structure are still limited; airborne lidar over the Alps found aerosol layers that did not follow the topography (Nyeki et al., 2000), and soundings in deep Alpine valleys show a mixed layer growing to about the height of the surrounding ridges in the afternoon (Chemel and Chollet, 2006). How strongly the elevated terrain heats the atmosphere in place is a related question: a valley warms more than the plain for the same sunshine because it has less air to heat per unit of surface (Steinacker, 1984; Whiteman, 2000; Wagner et al., 2015), and the heating drives the slope and valley winds and the export of valley air to the free troposphere (Henne et al., 2004; Schmidli, 2013; Leukauf et al., 2016). Numbers for these processes come from a handful of campaigns such as the Riviera project (Rotach et al., 2004), and from models (Weigel et al., 2006; Leukauf et al., 2016).

The measurement rests on a property of the logs that needs no polar and no segmentation: most flight recorders write two altitudes, one from a barometer and one from a GNSS receiver. Together they give the virtual temperature of the air between two pressure levels, layer by layer, through the hypsometric equation. Each ingredient has precedents, and none of them covers the valley atmosphere. Temperature from paired geometric and pressure altitudes is established for airliners tracked by Mode-S radar and ADS-B (de Haan, 2011; de Leege et al., 2013; Stone and Kitchen, 2015; de Haan, 2026), at cruise level and along airport approaches over flat terrain, not between 1.5 and 4 km inside mountain valleys. Instrumented paragliders have measured temperature and humidity directly (Coppola, 2021), on a handful of flights. Large archives of flight logs have been read for the structure, strength and ceiling of thermals (Ultsch, 2012; Hernández-Aguayo et al., 2026) and for the evaluation of thermal forecasts (Nardon et al., 2026), and flight trajectories have supported empirical convection modelling (Pálenik et al., 2021); none of these extracts temperature. Crowdsourced pressure observations from smartphones give surface pressure (Mass and Madaus, 2014), and the GPS tracks of soaring birds give wind and thermals (Treep et al., 2016), not profiles. Dedicated valley campaigns such as Riviera and CROSSINN resolve the thermal structure of a valley with soundings, aircraft and remote sensing (Rotach et al., 2004; Chemel and Chollet, 2006; Adler et al., 2021), over weeks. What is new here is the combination: a multi-year record of valley temperature profiles, resolved by height and by hour, obtained from recreational flight logs alone on 437 flying days, and validated against an operational radiosonde. A paraglider climbs at 1–2 $\mathrm{m\,s^{-1}}$, along slopes and over valleys, for hours, and on a good day hundreds of them do it at once; the archive turns this into a repeatable observing system that needs no additional airborne sensor.

The paper asks three questions. Q1: Do the paired pressure and GNSS altitudes of routine paraglider logs yield layer-mean virtual temperatures of a mountain valley that agree with an independent radiosonde, and which screening of flights and instruments does this require? The answer decides whether the archives can serve as an observing system for the valley atmosphere. Q2: How much does the atmosphere of the Valais warm between morning and afternoon as a function of height, and how does this compare with the heating of Alpine valleys reported from campaigns and models? The answer bears on the valley heat budget and on the export of valley air to the free troposphere. Q3: Does the ceiling of thermal lift follow the terrain or become level during the day, at horizontal scales of 2.5 to 10 km? This is the terrain-following versus level boundary-layer top of De Wekker and Kossmann (2015), here observed within single days. Sections 2 and 3 present the method and its comparison with the radiosonde and Section 4 the checks that a flight is a usable hypsometer (Q1); Section 5 answers Q2 and Section 6 Q3; Section 7 discusses what a hypsometer flown along a slope measures and the observational coverage of the profiles.

2 Data and method

2.1 Flight logs

The source dataset contains 86 369 IGC flight logs of the Valais, 2007–2026, provided by XContest (XContest, 2026) under a confidentiality agreement; no raw log is redistributed and no pilot is identified. Instrument screening identifies 31 677 logs from dedicated flight instruments with distinct pressure and GNSS altitudes. Excluding smartphone applications and unnamed instruments, and requiring at least 20 candidate flights per day, gives 14 918 candidate logs on 437 days from 2011 onwards.

Quality control removes 22 duplicate recordings with identical B-record position and altitude sequences, retains valid fixes with strictly increasing timestamps, and requires at least 300 retained fixes per flight. Flights lie entirely inside the terrain mosaic (45–47° N, 6–9° E). The resulting 13 520 flights provide eligible layer estimates on 437 days, of which 392 meet the ten-flight threshold for at least one broad-window layer (Table 1). Sample sizes are reported separately for each contrast. The thermal-ceiling analysis uses the climb-selection criteria described in Section 2.4.

Table 1: Selection of the flight logs. Each row gives the criterion and what remains after it.

Step	Criterion	Remaining
Source	IGC logs of the Valais uploaded to XContest, 2007–2026	86 369 logs
Two altitudes	dedicated flight instrument with distinct pressure and GNSS altitudes	31 677 logs
Instrument and day	no smartphone application, no unnamed instrument; at least 20 candidate flights on the day; 2011 onwards	14 918 logs, 437 days
Fix quality	no duplicate recording; strictly increasing timestamps; at least 300 retained fixes; entirely inside 45–47° N, 6–9° E	13 520 flights, 437 days
Layer estimates	at least ten flights in a day, window and 25-hPa layer; recalibration rule of Section 2.3	392 days with at least one broad-window layer
Thermals	climbs identified by the rule of Section 2.4, 10–16 UTC	53 320 thermals, 386 days

2.2 Layer temperature from two altitudes

The retrieval proceeds in six steps for every flight.

1. *Pressure from pressure altitude.* The barometric altitude z_p written to the log refers to the standard atmosphere and is inverted to pressure, $p = 1013.25(1 - 2.25577 \cdot 10^{-5} z_p)^{5.25588}$ hPa. No QNH or calibration of the instrument enters; only changes of $\ln p$ along the flight are used.

2. *Time windows.* The fixes of a flight are assigned to the broad windows 07–10, 10–14 and 14–18 UTC, and separately to single hours for the hourly analysis. A flight-window must contain at least 100 fixes.

3. *Pressure layers.* Within each flight-window the fixes are grouped into 25-hPa layers between 875 and 525 hPa (layer centres 862.5 to 537.5 hPa, about 1300 to 5000 m). A layer is used when it holds at least 40 fixes that span at least 15 hPa.

4. *Regression.* For a hydrostatic column the local hypsometric relationship is

$$T_v = -\frac{g}{R_d}\left(\frac{\partial \ln p}{\partial z}\right)^{-1}, \tag{1}$$

with $g = 9.80665\,\mathrm{m\,s^{-2}}$ and $R_d = 287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$. In each layer, GNSS altitude z is regressed on $\ln p$ by least squares and the slope is multiplied by $-g/R_d$. This gives one virtual temperature per flight, window and layer.

5. *Screening.* Flight-windows whose layer values reproduce the standard atmosphere are flagged and removed by the recalibration rule of Section 2.3.

6. *Profile of the day.* The value of a day, window and layer is the median over the flights, with at least ten contributing flights. Its sampling uncertainty is obtained by resampling the flights (Section 4).

Figure 1 shows the steps on one flight. The result is an effective layer estimate along the sampled path; its equality to a local column temperature requires horizontal and temporal pressure variations to be negligible or suitably controlled (Section 7). GNSS geometric height is used as an approximation to geopotential height. A constant additive GNSS-height offset cancels in the slope. A constant additive pressure error does not cancel exactly: if $p_m = p + b$ in a vertical isothermal column, the inferred local temperature is multiplied by $(p + b)/p$. Similarly, a constant pressure-altitude offset is not an exactly constant $\ln p$ offset after the nonlinear inversion. Time-dependent calibration, pressure changes and horizontal gradients can also affect the slope. No general cancellation of barometric errors is assumed.

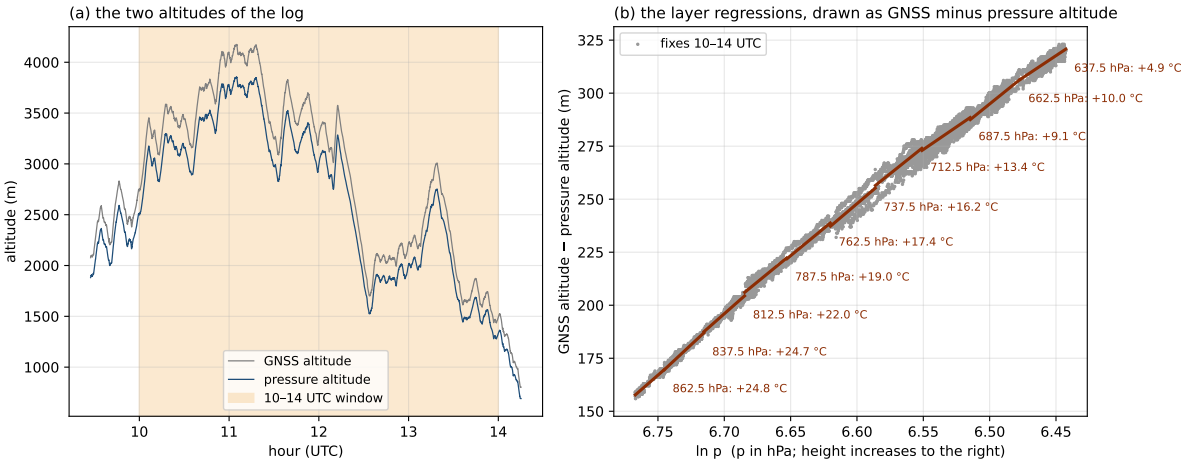


Figure 1: The method on one flight (Syride instrument, Valais, 20 August 2023). (a) Pressure altitude and GNSS altitude against time; the gap between them grows with height in warm air. (b) The fixes of the 10–14 UTC window, drawn as GNSS altitude minus pressure altitude against $\ln p$, with the least-squares line of each 25-hPa layer and the virtual temperature that the regression of GNSS altitude on $\ln p$ yields for that layer, Eq. (1). The steeper the line, the warmer the layer relative to the standard atmosphere: 24.8 °C at 862.5 hPa, 4.9 °C at 637.5 hPa.

The measured thermodynamic variable is virtual temperature, $T_v \simeq T(1 + 0.608q)$ for unsaturated air, where T is air temperature in kelvin and q is specific humidity in $\mathrm{kg\,kg^{-1}}$. This is the temperature variable entering the hydrostatic equation. We use "hypsometric temperature" for this measurement and retain T_v in equations and quantitative comparisons. Changes satisfy $\Delta T_v = \Delta T + \Delta(T_v - T)$. Section 5.1 quantifies the second term at Payerne to assess its regional scale. Sampling uncertainty and systematic sensitivities are evaluated in Section 4.

2.3 Instruments

Instrument diagnostics use samples selected for the comparisons below, including smartphone and pre-2011 records. They are distinct from the sample used for the temperature profiles.

Recalibration. Some instruments continuously adjust their barometric altitude to the GNSS altitude. Their layer temperatures reproduce the standard atmosphere whatever the real air. A flight-window with at least two layer estimates is flagged when the absolute median difference from the standard atmosphere is below 1.5 K. This is a temperature-dependent screening heuristic, not an independent identification of recalibration. For all terrain masks and hourly retrievals, the flag is inherited from the unmasked broad window, preventing the filter from being retuned separately to each result. Sensitivity to omitting the rule is reported with the analysis tables. Without the rule, the morning-to-afternoon increase at 787.5 hPa is 3.68 K (95 % interval 3.28–4.38 K; 112 days) instead of 3.55 K, and the median difference from Payerne is $+0.57$ K (1621 comparisons) instead of $+0.67$ K; the rule removes 6.03 % of eligible broad-window flight segments.

Smartphone logs. A diagnostic on 3 688 flights of 2015–2026 from days with at least three smartphone and three dedicated-instrument logs each (1 718 dedicated instruments, 1 233 XCTrack on Android, 730 Flyskyhy on iPhone; flights of at least 30 min and 500 m of altitude range) regressed the difference between pressure and GNSS altitude of each flight on altitude and on the altitude change of the preceding five minutes. In 82 % of the Flyskyhy logs and 15 % of the XCTrack logs the pressure altitude stays within 30 m of the GNSS altitude for the whole flight: the application ties the barometer to GNSS and the hypsometric signal is absent (dedicated instruments: 5 %). The untied XCTrack logs retain 88 % (95 % interval 85–94 %) of the temperature signal of the dedicated instruments on the same days, with slightly higher noise (2.6 against 2.2 m), and the tied fraction depends on the version of the application (12–34 % for versions 0.9.4–0.9.8 of 2020–2023; 9 % tied and 98 % of the signal retained for version 0.9.12 of 2025–2026). A thermal lag of the phone's sensor after climbs is small, -0.2 K per 500 m of climb (95 % interval -0.6 to $+0.2$ K). Because the log does not record the setting of the application, all smartphone logs are left out here; recent versions of XCTrack could be admitted with the recalibration rule above.

Early instruments. Profiles from 2008–2010, when the first generation of flight instruments dominated, lie 2–5 K below the radiosonde (medians -4.7, -3.6 and -2.1 K); from 2011 on the difference is stable between $+0.2$ and $+1.5$ K in every year. All results use 2011–2026.

2.4 Thermals

For Section 6, climbs were identified in the same logs by the following rule, which is simpler than the segmentation of the companion paper and sufficient here, since only the top height of each climb is used.

1. The vertical velocity of the air is the barometric vertical speed of the glider, smoothed over 15 s, plus a constant sink of the glider of 1.3 m s^{-1}.
2. A fix is circling when the heading changes by more than 60° within a centred 20-s window.
3. A climb is a contiguous circling stretch of at least 120 s with at least 150 m of height gain; its top is the highest pressure altitude in the stretch (heights of thermals are pressure altitudes throughout; the offset between pressure altitude and true height, common to the flights of a day, largely cancels in the within-day deviations of Section 6).
4. Only climbs between 10 and 16 UTC on days from 2011 on are kept: 54 481 thermals on 401 days.

5. Days on which the thermal tops pile up at one height (95th minus 70th percentile of the tops below 150 m, a cloud base or an airspace limit) are excluded: 15 days, leaving 53 320 thermals on 386 days.

The thresholds of all steps of this section are collected in Table 2.

Table 2: Numerical thresholds and parameters used in the paper, with the section where each enters.

Quantity	Value	Section
Candidate flights per day	≥ 20	2
Retained fixes per flight	≥ 300	2
Years	2011–2026	2.3
Broad windows	07–10, 10–14, 14–18 UTC	2
Pressure layers	25 hPa, 875–525 hPa	2
Fixes per flight-window	≥ 100	2
Fixes per layer regression	≥ 40, spanning ≥ 15 hPa	2
Flights per day, window and layer	≥ 10	2
Recalibration rule	$\lvert$median difference from standard atmosphere$\rvert < 1.5$ K over ≥ 2 layers	2.3
Bootstrap replicates	999 (flights), 2000 (days, contrasts), 4000 (energy proxy)	4
Spatial blocks	5-km cells combined with integer UTC hours	4
Cell-and-family control	10-km cell, ≥ 2 flights per cell, family and window	4
Terrain masks	radii 100, 300, 600, 1000, 1500 m; terrain $\geq$ flight height -50 m; free air additionally > 500 m above ground	4.3
Humidity comparison	Payerne $T_v - T$, 00 and 12 UTC, matched dates	5.1
Energy proxy	layers 800–700 hPa, dates with all four layers	5
Circling	heading change $> 60^\circ$ in 20 s	2.4
Climb	≥ 120 s, ≥ 150 m gain; air velocity = baro vertical speed (15 s) $+1.3\,\mathrm{m\,s^{-1}}$	2.4
Thermals used	10–16 UTC; days with piled-up tops (P95$-$P70 < 150 m) excluded	2.4
Ceiling cells	2.5, 5, 10 km; ≥ 8 thermals per cell; ≥ 4 cells per day; ceiling = 90th percentile of tops	6
Remaining-climb check	cuts at 3.0, 2.5, 2.0, 1.5 $\mathrm{m\,s^{-1}}$; trend over previous 300 m; climbs < 325 m keep their top	6.1

2.5 Use of AI tools

The author conceived the study, obtained the data, designed the analyses and interpreted the results. AI assistants (Claude, Anthropic; Codex, OpenAI) were used as programming aids, for an independent recomputation of the temperature analysis from the raw logs and of the thermal-ceiling analysis from the archived derived data, which reproduced the original results before the quality screening was tightened, and for language editing. The author checked all results and takes full responsibility for the content.

3 Comparison with the Payerne radiosonde

Payerne (491 m, on the Swiss Plateau, about 100 km from the centre of the Valais) launches radiosondes at 00 and 12 UTC. The IGRA2 data (Durre et al., 2018) supply virtual temperatures

averaged over the same 25-hPa layers. We compare the 10–14 UTC profiles with the 12 UTC sounding on 381 days, giving 2297 layer comparisons.

Between approximately 2200 and 3600 m (layer centres 787.5–662.5 hPa), the median flight-minus-sonde difference is +0.67 K and its scaled median absolute deviation is 1.47 K across 1545 comparisons on 370 days (Fig. 2). The scaled MAD is 1.4826 median$|x - \text{median}(x)|$ and describes dispersion; it is not a confidence interval or a median standard error.

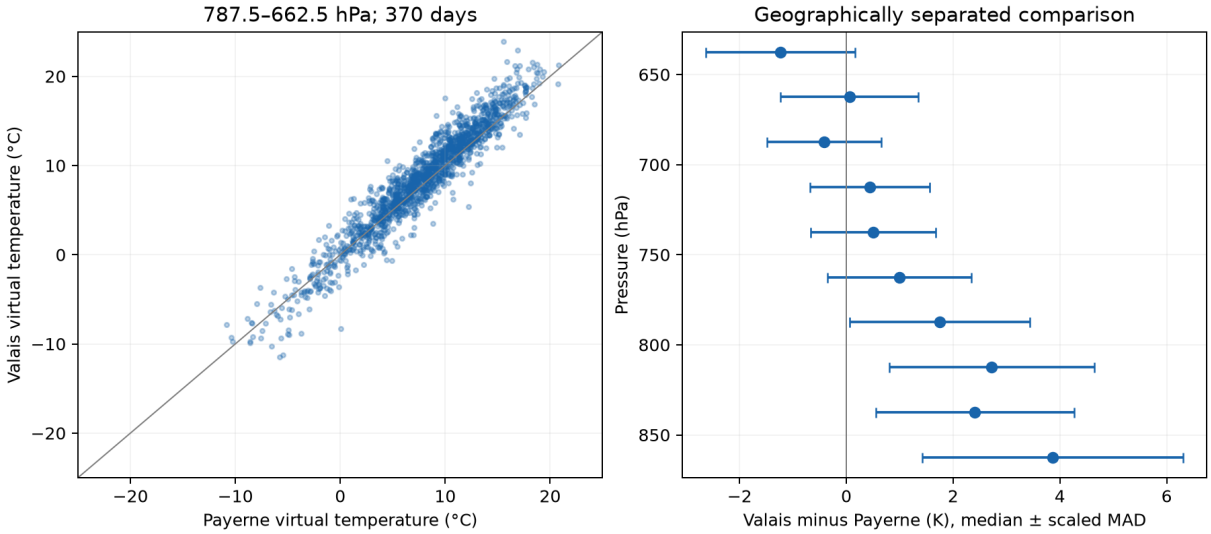


Figure 2: Quality-screened flight estimates compared with Payerne at 12 UTC. Left: layer temperatures at 787.5–662.5 hPa. Right: median difference and scaled MAD by layer. Differences combine instrumental, sampling and geographical effects.

The lower-layer positive differences are consistent with enhanced daytime temperatures over the valley. Because Payerne samples a different location and potentially a different air mass, this comparison measures regional agreement rather than local absolute accuracy. The within-day contrasts below test the valley's daytime evolution independently of this spatial difference, and the humidity analysis assesses the magnitude of the regional moisture contribution.

4 Uncertainty and robustness checks

4.1 Sampling uncertainty and dependence

Three resampling schemes are used, each matched to the unit that is treated as independent (Hesterberg, 2015).

1. *Flights within a day, window and layer.* The sampling error of a daily/window/layer median is obtained by drawing the contributing flight estimates with replacement (999 replicates) and taking the median of each draw; the standard error is the standard deviation of these medians, the interval its 2.5th and 97.5th percentiles. Fixes are never resampled individually. Across 4114 eligible groups, the median standard error is 1.34 times scaled-MAD/$\sqrt{n}$, which shows why the sampling uncertainty of the median is estimated directly rather than from a formula.

2. *Spatial-temporal blocks.* To check that flights are not strongly dependent through place and time, the flights of a group are also resampled as blocks of 5-km cells combined with integer UTC hours, each block drawn with all its flights. Across 4010 groups with at least five blocks, the median ratio of the block standard error to the flight standard error is 1.00.

3. *Days for contrasts.* A morning-to-afternoon or hour-to-hour contrast is first formed within each day and layer (afternoon minus morning median of that day), and the days are then resampled (2000 replicates); the reported interval is the 2.5th to 97.5th percentile of the median contrast over the draws. Calendar-week blocks provide a further dependence check and preserve the principal warming signals.

Reported intervals are pointwise 95 % sampling intervals; instrument and sampling sensitivities are assessed separately. Sparse cases are identified by their sample sizes.

4.2 Instrument and sampling sensitivities

Instrument-family comparisons on dedicated-instrument logs before the fix-level quality filters give differences of approximately 1.1 K between grouped families, with different absolute offsets against Payerne. A fixed family offset can cancel in a within-day contrast only insofar as the contributing instruments and their errors remain comparable. We therefore additionally compare hours or broad windows within the same day, pressure layer, approximately 10-km cell and instrument family, requiring at least two flights per cell-family-window. Cell-family contrasts are reduced to one median per day before resampling days. These controls reduce geographical and family-composition differences but do not remove within-family calibration drift or guarantee equal spatial sampling.

A climb/sink comparison on the same diagnostic dataset gives midday differences of about -0.23 K. It remains a useful complementary diagnostic of direction-dependent errors, without proving that pressure tendency or sensor drift has been uniquely separated. No single offset such as ± 0.5 K is treated as a comprehensive systematic-error bound.

4.3 Pointwise exclusion near terrain

The terrain mask is applied to individual fixes *before* each regression. SRTM 3-arc-second terrain (Farr et al., 2007) is searched within horizontal radii of 100, 300, 600, 1000 and 1500 m for terrain reaching at least the GNSS flight height minus 50 m. The operational distance is therefore horizontal distance to terrain of similar elevation, not slope-normal distance. Tests use approximately 64-m east–west and 92.6-m north–south grid spacing. All terrain-selected fixes must have non-negative height above valid terrain. The strict free-air mask also requires more than 500 m above ground. Grid resolution and GNSS/geoid differences limit interpretation of the 100-m class.

After masking, each regression must still contain at least 40 fixes spanning 15 hPa. Profiles are compared on the same dates, windows and layers. A second diagnostic pairs the masked and unmasked retrievals from the same flights, separating retrieval changes from changes in flight selection. All masks use the same recalibration flags.

At 787.5 hPa (about 2200 m), removing fixes within 300 m of terrain leaves a morning-to-afternoon increase of 3.20 K (95 % interval 2.57–4.00 K; 74 days). The paired change of warming relative to the unmasked result on exactly those days is -0.06 K (-0.33 to +0.21 K; 74 days). The warming is therefore not explained by the inclusion of the nearest-terrain fixes in this sample. Stricter exclusion is much more costly: the free-air selection retains only one paired day at 787.5 hPa, compared with 12 at 762.5 hPa and 35 at 712.5 hPa. Consequently it cannot establish lower-layer morning robustness. The latter two free-air warming estimates are 2.85 K (95 % interval 2.07–3.99 K; 12 days) and 1.67 K (95 % interval 1.03–2.24 K; 35 days), respectively.

Near-only versus free-only comparisons also remain sampling limited. At 762.5 hPa at midday, the near-100-m minus free-air contrast is -0.35 K (-1.61 to +0.05 K; 27 days). This comparison resolves no clear positive near-slope temperature excess. It does not establish the

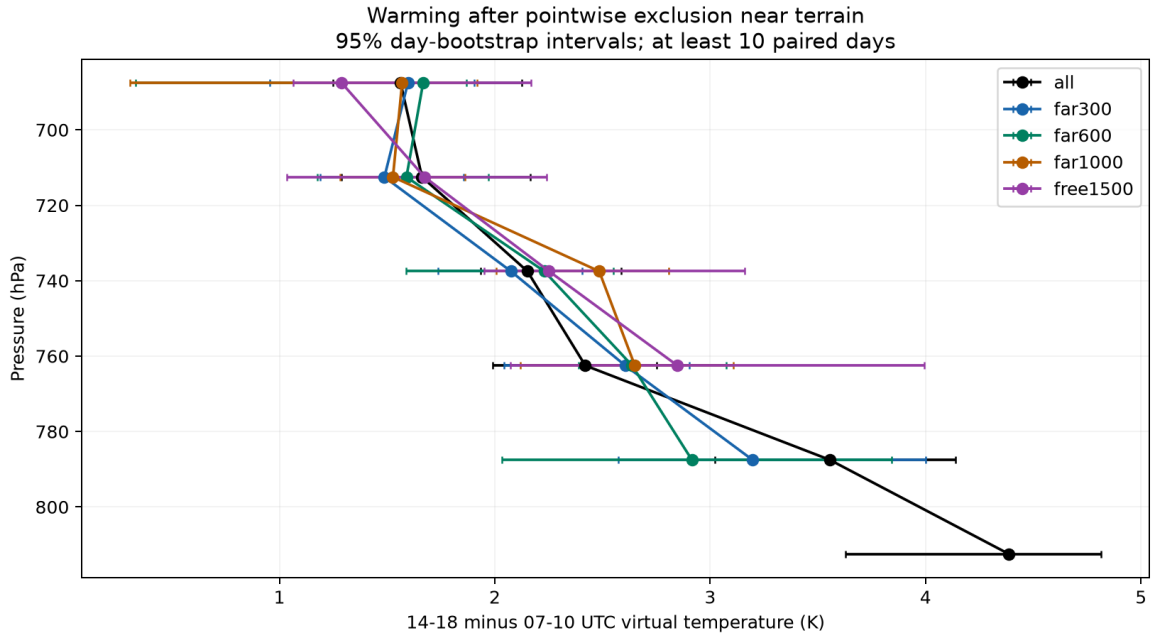


Figure 3: Morning-to-afternoon virtual-temperature changes after pointwise terrain exclusion. Each curve uses dates with at least ten flights in both windows for its mask; different masks therefore have different date sets. Intervals resample paired daily differences. Matched-date changes are reported separately in the text and tables. Only groups with at least ten paired days are plotted. "far300/600/1000" denote the exclusion radius in metres; "free1500" additionally requires 500 m above ground.

temperature of a local slope layer, because the two samples also differ in their locations and flight paths. Section 7 develops a physical explanation for the sign and a flight-path comparison to test it.

5 Observed diurnal evolution

The broad-window estimates show morning-to-afternoon virtual-temperature increases of 4.39 K (95 % interval 3.63–4.82 K; 28 days) at approximately 1940 m, 3.55 K (95 % interval 3.02–4.14 K; 107 days) at 2200 m, 2.42 K (95 % interval 1.99–2.75 K; 108 days) at 2470 m, 2.15 K (95 % interval 1.93–2.59 K; 86 days) at 2740 m and 1.66 K (95 % interval 1.29–2.16 K; 45 days) at 3020 m. These are medians of 14–18 minus 07–10 UTC contrasts within individual days. Their intervals express sampling uncertainty across the observed flying days, not all-weather climatological uncertainty.

Hourly regressions use the same pressure spans and minimum numbers of fixes; each hourly profile still requires ten flights. Figure 4 compares each observed hour with 10–11 UTC on the same dates. At 787.5 hPa the median contrasts are -2.02 K at 08–09 UTC (23 paired days), -1.03 K at 09–10 UTC (61), +0.92 K at 11–12 UTC (46), +1.31 K at 12–13 UTC (56), +2.44 K at 13–14 UTC (49), +2.38 K at 14–15 UTC (48) and +2.97 K at 15–16 UTC (27). Each point uses its available date intersection with the reference hour; it is not a single fixed-cohort trajectory. Restricting 09–16 UTC to the same complete dates leaves 14 days at 787.5 hPa and retains the rising evolution. The corresponding complete-date tables are supplied.

Matching the 10-km cell and instrument family retains a broad-window increase of 3.26 K (0.66–4.85 K; 63 days) near 2200 m. These controls reduce changes in spatial and instrument composition. Together with the fixed-date hourly comparison and pointwise terrain screening, they support a daytime warming signal extending beyond the nearest-slope measurements. The resolved evolution begins with the first sufficiently sampled hour; no extrapolation to dawn is required.

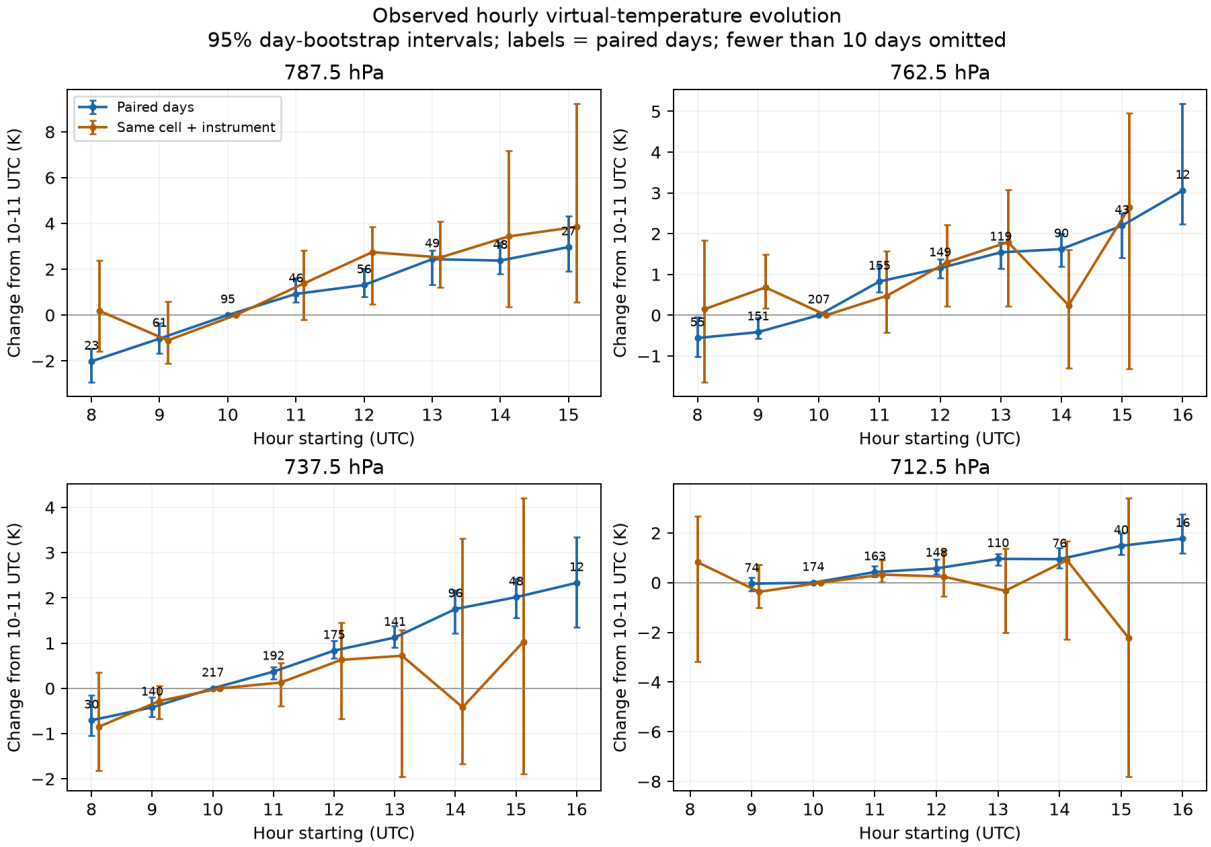


Figure 4: Observed hourly virtual-temperature contrasts relative to the measured 10–11 UTC hour, paired within days. Blue: profiles with at least ten flights per hour; labels give paired-date counts. Orange: additionally matched 10-km cell and instrument family, at least two flights in each cell-family-hour, summarised once per day. The orange and blue samples need not contain the same dates. Pointwise 95 % intervals resample days. Groups with fewer than ten dates are omitted. The reference hour is identically zero by construction.

How do these numbers compare with what is known about the heating of Alpine valleys? The decrease of the warming with height, from more than 3.5 K at 2200 m to 1.7 K at 3000 m, is the signature of a valley atmosphere heated from below and from its slopes, with the volume effect of Steinacker (1984) and Wagner et al. (2015): the same sunshine heats less air in a valley than over the plain. Observations in deep valleys, in the Ina valley by Kuwagata and Kimura (1995) and in the Alps by Chemel and Chollet (2006), describe a daytime warming of the whole valley atmosphere up to about the height of the surrounding ridges, and the Riviera heat budget of Weigel et al. (2006) attributes it to the turbulent heat flux from the slopes together with subsidence over the valley centre, which brings warm air down from above. The profiles here do not separate these contributions; what they add is the vertical structure of the result on hundreds of days: a warming that is largest in the lowest layers and reaches, diminished, the height of the ridges, as the idealised simulations of Leukauf et al. (2016) and the campaign observations describe for single days. The column-equivalent proxy of Section 5.2, 2.5 MJ m^{-2} for 800–700 hPa, corresponds to three hours of a surface sensible heat flux of about 230 W m^{-2} and is therefore compatible in scale with these budgets, without closing one.

5.1 Humidity contribution on matched observing days

The Payerne soundings contain both T and T_v in the same pressure layers. We calculate $C = T_v - T$ for each sounding, then difference C within each paired date and layer. Only dates with both morning and afternoon Valais estimates and both relevant Payerne soundings are retained. This directly quantifies the moisture contribution to the Payerne temperature change. Table 3 gives medians and interquartile ranges for 00–12 UTC, alongside Valais contrasts on exactly those dates. Specific humidity inferred from the ratio of the layer-mean temperatures is an effective layer value, $q \simeq (T_v/T - 1)/0.608$; the correction C itself requires no such inversion.

Table 3: Payerne humidity contribution on dates with paired Valais morning and afternoon estimates. ΔC is the 12 minus 00 UTC change in $T_v - T$; brackets give the interquartile range across days, not a confidence interval. The last column gives the Valais 14–18 minus 07–10 UTC contrast on the same dates. Each entry is the median of paired differences.

Layer (hPa)	Days	Δq (g kg^{-1})	ΔC (K) [IQR]	Valais ΔT_v (K)
812.5	24	−1.64	−0.28 [−0.40, −0.05]	4.50
787.5	96	−0.76	−0.13 [−0.29, +0.07]	3.73
762.5	96	−0.36	−0.06 [−0.28, +0.08]	2.39
737.5	75	−0.72	−0.12 [−0.29, +0.06]	2.16
712.5	39	−0.13	−0.03 [−0.15, +0.11]	1.50

Median moisture contributions range from −0.28 to −0.03 K for 00–12 UTC; the complementary 12–00 UTC next-day comparison gives +0.05 to +0.22 K. Thus the regional median moisture contribution is only a few tenths of a kelvin, compared with several kelvin of valley warming. This independent evidence supports a predominantly thermal interpretation of the observed signal. Payerne supplies a regional scale comparison, not a local humidity correction: its location and sounding intervals differ from those of the flights, and the median changes do not bound individual days or intermediate hours.

5.2 Column-equivalent warming energy

As a descriptive column-equivalent energy proxy, we compute

$$E^* = \frac{c_p}{g} \sum_{800\text{–}700\,\mathrm{hPa}} \Delta p \, \Delta T_v, \tag{2}$$

using $c_p = 1005\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $\Delta p = 2500\,\mathrm{Pa}$ and only dates with paired estimates in all four layers. The median is $2.50\,\mathrm{MJ\,m^{-2}}$ (95 % sampling interval 2.41–2.95; 43 days; 4000 bootstrap replicates). Summing different layers' medians on different day sets is avoided. This proxy is not a measured valley-integrated heat-storage budget: virtual temperature includes humidity, the flight medians are not area-weighted means, pressure surfaces intersect mountain terrain, and advection and moving sampling volumes are not resolved. It is consistent in scale with substantial daytime warming, but does not close the heat budget discussed in mountain-boundary-layer studies (Rotach et al., 2004; Weigel et al., 2006; Schmidli, 2013).

6 The ceiling of thermals and the terrain

6.1 Thermals are left early

Only 2.5 % of the thermals are flown out, that is, left with a climb rate below $1\,\mathrm{m\,s^{-1}}$; the median climb rate at the moment of leaving is $2.0\,\mathrm{m\,s^{-1}}$. The height at which a thermal is left is therefore the pilot's decision, and the top of the lift is a lower bound. To see how much is missing, a remaining-climb function was learned from the flown-out thermals: each is cut artificially at the moment its climb rate first drops to 3.0, 2.5, 2.0 or $1.5\,\mathrm{m\,s^{-1}}$, and the height it still gained after that is recorded by the trend of the climb over the previous 300 m and the strength of the thermal. Learned on the flown-out thermals of half of the days and tested on the other half, the function predicts the remaining climb with a median error of +25 m and a spread of 150 m. Of the 51 978 thermals left early, 18 680 gained less than 325 m, too little for the trend over the last 300 m, and keep their observed top; of the 33 298 others, 10 655 fall in classes whose learned remaining climb is zero (the climb was already fading), and 22 643 receive a remaining climb, 225 m in the median (quartiles 150–475 m). The function is learned from the 2.5 % of thermals that pilots flew out, which may differ from the rest; this is the reason for treating the correction as a check and the observed top as the quantity.

6.2 Does the ceiling follow the terrain?

What is measured is the upper envelope of the observed climbs, called the ceiling here for brevity. It is a lower bound on the tops of the thermals themselves; whether those lie above or below the top of the mixed layer depends on its definition and on the overshoot of thermals into the stable air, which these data do not resolve. For each day and each cell of a grid of 2.5, 5 or 10 km with at least eight thermals, the ceiling is the 90th percentile of the thermal top heights, and the terrain height of the cell is the mean of the SRTM 3″ elevation (Farr et al., 2007). Within each day with at least four such cells, ceiling and terrain are expressed as deviations from the day's mean, so that the overall level of the day cancels, and the ceiling deviation is regressed on the terrain deviation over all days (Fig. 5). The 95 % intervals come from resampling the days. Within a day the terrain of the cells spans 750 m (2.5 km cells) to 1220 m (10 km) in the median, and the ceiling lies at 3230–3290 m.

The factor is 0.32 (95 % interval 0.23–0.42) at 2.5 km, 0.21 (0.15–0.28) at 5 km and 0.22 (0.17–0.25) at 10 km, from 925, 1399 and 1650 cell-days on 91, 143 and 184 days. A cell whose terrain lies 1000 m above the day's mean has a ceiling 210–320 m higher, not 1000 m. With the remaining climb of Section 6.1 added to every thermal left early, the factors are 0.23 (0.18–0.29) at 5 km and 0.21 (0.17–0.26) at 10 km, against 0.20 and 0.21 for the same cell-days without it: this particular correction changes the fitted factor little on the matched sample; other selection effects remain possible. The direction of any remaining selection bias cannot be established from these data alone.

The cells of a day are not all flown at the same hour, and the ceiling rises during the day, so

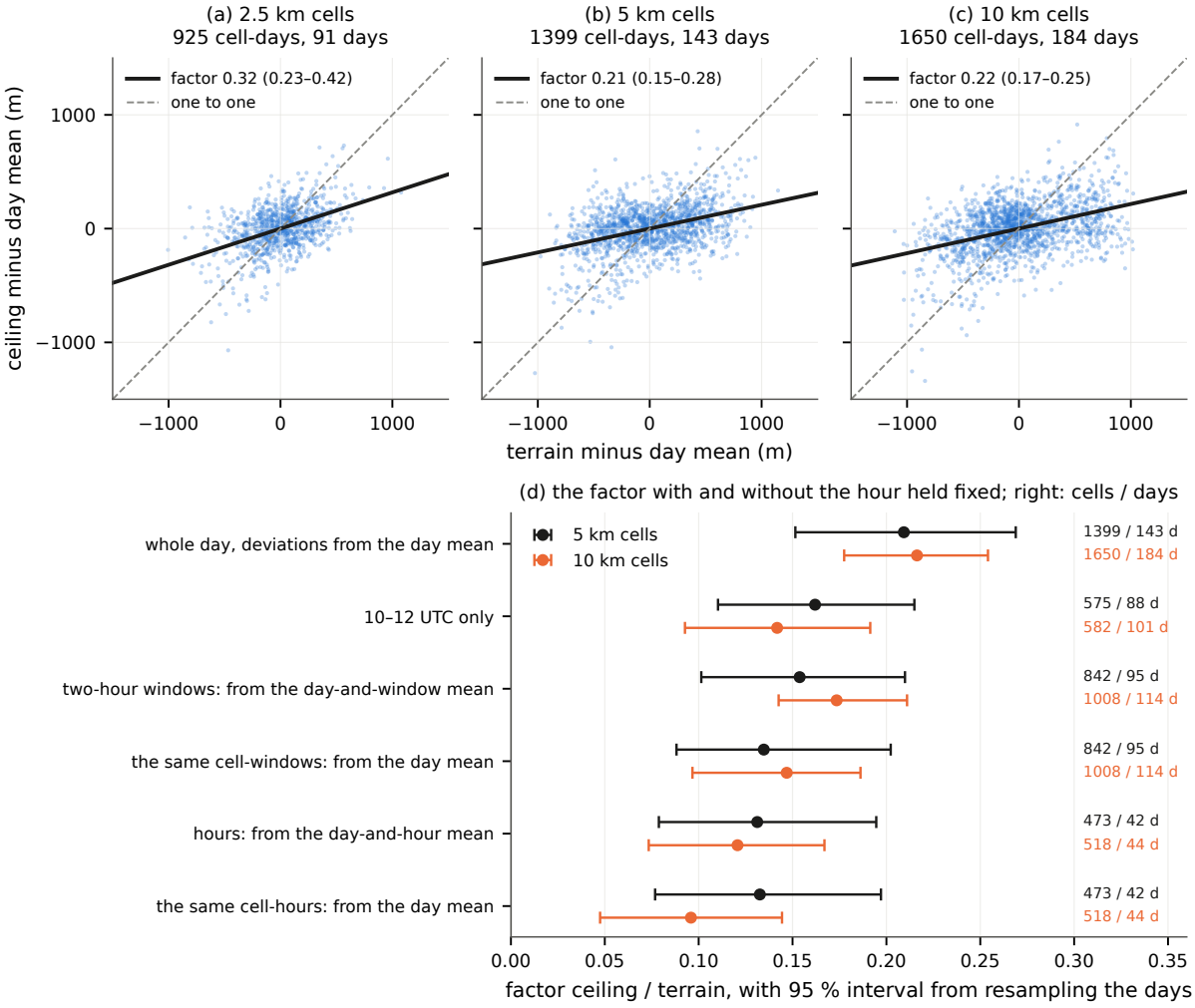


Figure 5: Ceiling of thermal lift against the terrain, within the same day. Each point is one cell on one day: the 90th percentile of the thermal tops in the cell, minus the day's mean over its cells, against the mean terrain height of the cell, minus the day's mean. (a)–(c) Cells of 2.5, 5 and 10 km. The line is the fit, with its factor and 95 % interval from resampling the days; the dashed line is a ceiling that follows the terrain one to one. (d) The factor for the whole day, for 10–12 UTC alone, and for the samples that allow the hour of the day to be held fixed: two-hour windows and single hours, each once with the deviations from the mean of the day and window or hour, and once, on exactly the same cells, from the mean of the day. Right: number of cell-days, cell-windows or cell-hours, and of days.

the factor could contain a time-of-day effect. Within a day the ceiling rises by 77 m per hour (90th percentile of the tops per day and hour, 773 day-hours on 169 days; −170 m at 10 UTC, +200 m at 15 UTC relative to the day's mean), and the thermals over the lowest cells are flown 0.4–1.1 h later than those over the highest (12.3–12.5 h against 11.4–11.9 h). This gives the lowest cells a late-day bonus of 33–85 m, which lowers the apparent factor by at most 0.03–0.08; a correction for the hour should raise it slightly. Figure 5d shows the test. Taking cell-hours with at least eight thermals in the hour and at least four cells per day and hour, the factor from the deviations from the mean of the day and hour is 0.13 (0.08–0.19) at 5 km and 0.12 (0.07–0.17) at 10 km, from 473 and 518 cell-hours on 42 and 44 days; from the deviations from the day mean, on exactly the same cell-hours, it is 0.13 (0.08–0.20) and 0.10 (0.05–0.14). With two-hour windows (10–12, 12–14, 14–16 UTC) it is 0.15 (0.10–0.21) and 0.17 (0.14–0.21) from 842 and 1008 cell-windows on 95 and 114 days, against 0.13 (0.09–0.20) and 0.15 (0.10–0.19) on the same cell-windows from the day mean; within 10–12 UTC alone it is 0.16 (0.11–0.21) and 0.14 (0.09–0.19). Holding the hour fixed thus changes the factor by +0.00 to +0.02, the direction and size expected from the rise of the ceiling; the lower values of the hourly samples against the day-wide 0.21–0.22 come from the cells and days these samples select, the busy ones, not from the hour.

The upper envelope of the climbs over the Valais between 10 and 16 UTC is therefore nearly level, with a slight doming over high terrain, on all scales from 2.5 to 10 km: a factor of 0.2–0.3 across the day, 0.10–0.17 in the samples that allow the hour to be held fixed, and less than 0.1 of it from the hour. This is the afternoon state that De Wekker and Kossmann (2015) describe as reached when the valleys heat fast enough, and it is consistent with the lidar observations of Nyeki et al. (2000); here it is measured on 184 days. The flight-top envelope and the Payerne comparison do not uniquely identify the physical mixed-layer top. The thermal ceiling is one of the three observables that Hernández-Aguayo et al. (2026) relate to reanalysis predictors over France; the present analysis quantifies its relationship to terrain within the same day.

7 What a hypsometer flown along a slope measures

Under hydrostatic balance, pressure measures the weight of the air column overhead per unit area. The hypsometric method uses the height difference between pressure levels to measure the temperature of the intervening air: warm air occupies a greater depth for the same pressure ratio. A vertical climb samples successive levels of one column. A climb along a slope also moves between columns. The pressure differences between those columns can reinforce or offset the signal from the warm air through which the pilot flies.

Consider a thin warm layer lying along a planar slope. To isolate the effect of flight geometry, suppose that the layer is stationary, its temperature excess depends only on distance normal to the slope, the background temperature is constant, and pressure approaches the same background atmosphere well above the layer. These assumptions specify a simple hydrostatic thought experiment. A pilot climbing vertically moves out through the warm layer, and the pressure change with height records its elevated temperature. A pilot climbing parallel to the slope at constant distance remains at the same position within the warm-layer structure.

In the parallel case, the warm air overhead produces the same proportional pressure deficit relative to the background at each point along the climb. This constant proportion cancels when the pressure ratio between two points is taken. The retrieved temperature therefore equals the background temperature, even though the pilot encounters warmer air. The important condition is the unchanged proportional pressure deficit along the path, rather than proximity to terrain alone. This example shows how a warm layer can escape a hypsometric measurement along a particular path. It specifies a pressure field in vertical hydrostatic balance; it does not describe the full circulation driven by the layer or imply that real slope layers are generally invisible.

A changing pressure deficit gives a second possibility. If the proportional deficit becomes larger as the pilot climbs, pressure falls more rapidly along the path than it does in the background atmosphere. The retrieval then gives a lower temperature, even when the air encountered is warm. An increase in the strength of the warm layer uphill can produce this behaviour under the same upper-pressure condition. A weakening deficit has the opposite effect. Layer growth during the flight, changes in thickness and departures from a slope-parallel path must therefore be assessed through their effect on the pressure field; warming near the ground alone does not determine the sign of the retrieved anomaly.

This gives a physical interpretation to test against the near-slope observations. At 762.5 hPa, the midday near-100-m minus free-air contrast is -0.35 K (95 % interval -1.61 to $+0.05$ K; 27 days). Its sign is consistent with a pressure deficit strengthening along an ascending path. The interval includes zero, however, and the two samples differ in location and flight geometry; variable slope-layer temperatures and GNSS errors near terrain can also contribute. The contrast therefore motivates the mechanism as a hypothesis. Establishing it requires knowing whether a warm layer was present and how the sampled paths crossed its pressure field.

The logs provide a concrete way to investigate this hypothesis: compare climbs that leave a slope approximately vertically with climbs that follow it at nearly constant normal distance, matched in location, height, time and instrument. In the stationary uniform-layer example, the former detect warmth and the latter return the background temperature; a strengthening pressure deficit can make the latter appear cooler. This comparison requires slope-normal distance and path orientation at individual fixes. The horizontal terrain-distance mask used here does not by itself identify these paths. Colocated temperature and pressure observations would establish the actual warm-layer anomaly and test whether the predicted contrast occurs.

The existing terrain-exclusion test answers the broader observational question. Near 2200 m, removing fixes within 300 m of terrain preserves the morning-to-afternoon warming; on matched dates the change is only -0.06 K (95 % interval -0.33 to $+0.21$ K). Together with the cell-and-instrument controls, this shows that the daytime signal extends beyond the nearest-slope measurements. Its persistence does not depend on adopting the slope-layer explanation.

Coverage sets another limit. Pilots rarely sample stable air above the thermal ceiling, and the highest midday profile layer typically lies below the observed thermal tops. The profiles therefore describe the sampled valley atmosphere; they do not determine the unsampled stratification above the climbs or the overshoot of thermals into it.

8 Conclusions

Q1, the method. Routine paraglider flight archives provide a new observational record of valley temperature. With the screening of Section 2, dedicated instruments only, at least 20 flights on the day, the fix-quality rules and the recalibration rule, the paired pressure and GNSS altitudes of the logs yield layer-mean virtual temperatures that agree with the Payerne radiosonde, 100 km away, to $+0.67$ K in the median with a scatter of 1.47 K between 2200 and 3600 m on 370 days. Daily profiles exist for 392 days and hourly contrasts resolve the daytime evolution; resampling of flights and days quantifies the sampling uncertainty. The archives can therefore serve as an observing system for the valley atmosphere, with the limits set by where and when pilots fly.

Q2, the warming. Between morning and afternoon the atmosphere of the Valais warms by 3.55 K near 2200 m and 1.66 K near 3020 m, in the median over 107 and 45 days, and the hourly contrasts show the rise through the day. The signal persists after pointwise removal of near-terrain fixes and after matching spatial cells and instrument families. The median humidity contribution in matched Payerne soundings is a few tenths of a kelvin, supporting a predominantly thermal interpretation of the multi-kelvin changes. The 800–700-hPa column-

equivalent warming proxy is 2.50 MJ m^{-2}; interpreting it as a valley heat-storage budget would additionally require local humidity and spatial weighting. The magnitude and the decrease with height agree with the valley heating described by campaigns and models; what the record adds is its vertical structure on hundreds of days.

Q3, the ceiling. The upper envelope of observed climbs rises by only 210–320 m per 1000 m of terrain at 2.5–10 km scales: within single days the thermal ceiling is nearly level, with a slight doming over high terrain, the afternoon state that De Wekker and Kossmann (2015) describe, here measured on 184 days. The envelope is a lower bound on the tops of the thermals and does not by itself identify the mixed-layer top.

The terrain-exclusion tests establish the persistence of the daytime warming beyond the nearest-slope measurements. A hydrostatic thought experiment shows how a uniform warm layer can cancel from a slope-parallel retrieval when its proportional pressure deficit remains constant along the path, and how a deficit that strengthens along the climb produces a negative retrieved anomaly; the observed near-slope contrast motivates a test of this mechanism with slope-following and slope-leaving climbs and colocated observations. Recent smartphone applications that write an untied pressure altitude would widen the archive further. The approach opens existing recreational-flight archives to repeated measurements of the thermal structure of mountain atmospheres.

Code and data availability

The derived data, scripts and aggregate tables of the analyses presented here are archived at Zenodo (Korner, 2026a). Raw IGC logs remain confidential under the agreement with XContest; reproducing the pointwise analysis requires authorised access to them and the SRTM terrain tiles.

Author contributions

JK conceived the study, obtained the data, designed and performed the analyses, prepared the figures and wrote the paper.

Competing interests

The author declares no competing interests.

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